EXPANDED INFORMATION CORPUS: TESLA VEHICLES

Comprehensive Technical Reference for RAG Systems

SECTION 1: COMPLETE VEHICLE LINEUP — DETAILED SPECIFICATIONS

1.1 Model S — Flagship Electric Sedan

Aspect	Details
Production Years	2012–present
Platform Generations	Gen 1 (2012–2016), Gen 2 (2016–2021), Gen 3 "Palladium" (2021–present)
Body Style	5-door liftback sedan, drag coefficient: 0.208 Cd (lowest of any production car at launch of Ge
Dimensions	Length: 197.7 in, Width: 78.2 in (with mirrors), Height: 56.3 in, Wheelbase: 116.5 in
Curb Weight	Long Range: 4,561 lbs, Plaid: 4,766 lbs

Variant Deep Dive

Model S Long Range (Current)

- **Powertrain:** Dual motor AWD (one front induction, one rear permanent magnet)
- **Horsepower:** ~670 hp (estimated)
- **0–60 mph:** 3.1 seconds (with 1-foot rollout)
- **Top Speed:** 149 mph
- **EPA Range:** 405 miles (on 19" Tempest wheels); 382 miles (on 21" Arachnid wheels)
- **Quarter Mile:** 10.9 seconds @ 124 mph
- **Battery:** 100 kWh nominal capacity (approx. 95 kWh usable), 450V architecture, 7,104 cells (2170 format in earlier versions, updated chemistry in current)
- **Charging:** Peak Supercharger rate: 250 kW; 10–80% in approximately 25 minutes

Model S Plaid

- **Powertrain:** Tri-motor AWD — one front induction motor, two rear carbon-wrapped permanent magnet motors

- **Horsepower:** 1,020 hp (combined)
- **0–60 mph:** 1.99 seconds (with 1-foot rollout) — requires "Drag Strip Mode" and preconditioned battery
- **Top Speed:** 200 mph (requires carbon ceramic brake kit option)
- **EPA Range:** 359 miles (19" wheels); 326 miles (21" wheels)
- **Quarter Mile:** 9.23 seconds @ 155 mph (production car record at launch)
- **Key Engineering Feature:** Carbon-sleeved rotors in rear motors prevent expansion at high RPM, enabling sustained high-speed operation
- **Track Package:** Available option including carbon ceramic brakes, zero-G wheels, track-focused brake fluid, and updated suspension tuning

Historical Variants (No Longer in Production)

- **Model S 60 / 60D:** 60 kWh battery, discontinued 2017
- **Model S 75 / 75D:** 75 kWh battery, discontinued 2018
- **Model S 100D:** 100 kWh battery, predecessor to Long Range
- **Model S P100D "Ludicrous":** 0–60 mph in 2.28 seconds (with rollout), 778 hp, "Ludicrous Mode" launch control, predecessor to Plaid
- **Model S P85 / P85D:** Performance variants with "Insane Mode" (2014–2015)

1.2 Model 3 — Mass-Market Compact Sedan

Aspect	Details
Production Years	2017–present
Generation	Pre-Refresh (2017–2023), "Highland" Refresh (2023–present)
Body Style	4-door compact sedan, drag coefficient: 0.219 Cd (pre-refresh), 0.219 Cd maintained in Highland wi improved aero wheels
Dimensions	Length: 185.8 in, Width: 72.8 in (with mirrors folded), Height: 56.8 in, Wheelbase: 113.2 in

Aspect	Details
Curb Weight	RWD: 3,862 lbs, Long Range AWD: 4,034 lbs, Performance: 4,065 lbs

Variant Deep Dive

Model 3 Rear-Wheel Drive (Highland)

- **Battery Chemistry:** LFP (Lithium Iron Phosphate) — sourced from CATL, manufactured in China and assembled in structural pack format (Giga Shanghai vehicles) or modular (other markets)
- **Battery Capacity:** 60–62 kWh nominal (varies by market)
- **Range:** 272 miles (18" wheels) to 305 miles (optimized configurations)
- **0–60 mph:** 5.8 seconds
- **Top Speed:** 125 mph
- **Charging:** Peak Supercharger rate: 170 kW (due to LFP voltage characteristics); 10–80% in approximately 25 minutes
- **Daily Charging Recommendation:** 100% (LFP chemistry allows and benefits from regular 100% charging for BMS calibration)
- **Drive Unit:** Single permanent magnet motor, rear axle, 239 hp, 310 lb-ft torque

Model 3 Long Range AWD (Highland)

- **Battery Chemistry:** Nickel-based (NCM 811 or similar high-nickel cathode)
- **Battery Capacity:** 75–82 kWh nominal (usable varies by market)
- **Range:** 341–358 miles (EPA estimates vary by wheel configuration)
- **0–60 mph:** 4.2 seconds
- **Top Speed:** 145 mph
- **Charging:** Peak Supercharger rate: 250 kW; 10–80% in approximately 20 minutes (optimized charge curve)
- **Daily Charging Recommendation:** 80–90% (for optimal battery longevity)
- **Drive Units:** Front induction motor (disengages at highway speeds for efficiency), rear permanent magnet motor (primary propulsion)
- **Combined Power:** 490 hp (estimated)

Model 3 Performance (Highland)

- **Distinguishing Features:** Sports seats with enhanced bolstering, carbon fiber spoiler (functional downforce), 20" forged wheels (optional), track-ready brake pads, aluminum alloy pedals

- **Powertrain:** Dual motor AWD with Performance-specific software calibration and inverter hardware
- **0–60 mph:** 2.9 seconds (with 1-foot rollout)
- **Top Speed:** 163 mph
- **Range:** 279–303 miles (EPA)
- **Track Mode V3:** Advanced vehicle dynamics control allowing adjustable torque split, stability control intervention levels, regenerative braking intensity, and cooling prioritization
- **Charging:** Same as Long Range (250 kW peak), but thermal management optimized for track use with "Track Mode Preconditioning" option

Pre-Refresh Model 3 (2017–2023) — Notable Variations

- **Standard Range Plus (SR+):** 54 kWh battery, 263 miles range, 0–60 in 5.3 seconds
- **Mid-Range:** Limited production (2018–2019), 62 kWh, 264 miles range
- **Long Range RWD:** Rare variant (2017–2018), 310 miles range, single motor, highly sought after for efficiency
- **Performance Stealth:** Unbadged Performance model with Long Range wheels and suspension, no spoiler, limited production

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1.3 Model X — SUV with Falcon Wing Doors

Aspect	Details
Production Years	2015–present
Generations	Gen 1 (2015–2021), Gen 2 "Palladium" (2021–present)
Body Style	5-door SUV, drag coefficient: 0.24 Cd (remains lowest for an SUV)
Dimensions	Length: 198.3 in, Width: 78.7 in (with mirrors folded), Height: 66.0 in, Wheelbase: 116.7 in
Curb Weight	Long Range: 5,185 lbs, Plaid: 5,390 lbs

Aspect	Details
Falcon Wing Doors	Double-hinged, ultrasonic sensors for obstacle detection, opening height clearance: 6' 8" minimum open in as little as 12 inches of side clearance

Key Variants

Model X Long Range

- **Powertrain:** Dual motor AWD
- **0–60 mph:** 3.8 seconds
- **Range:** 335 miles (EPA est., 20" wheels)
- **Towing Capacity:** 5,000 lbs (with factory tow package)
- **Seating Configurations:** 5-seat (standard), 6-seat (captain's chairs, $6,500 option), 7-seat (2+3+2 configuration, $3,500 option)

Model X Plaid

- **Powertrain:** Tri-motor AWD (identical architecture to Model S Plaid)
- **0–60 mph:** 2.5 seconds (with rollout) — world's quickest SUV at launch
- **Top Speed:** 149 mph (limited due to vehicle dynamics)
- **Range:** 326 miles (EPA)
- **Quarter Mile:** 9.9 seconds @ 140 mph
- **Seating:** 6-seat only in Plaid configuration

Historical Variants

- **Model X 75D:** 75 kWh battery, discontinued 2018
- **Model X 100D:** 100 kWh battery, predecessor to Long Range
- **Model X P100D Ludicrous:** 778 hp, 0–60 in 2.8 seconds (with rollout), "Ludicrous Mode"

Falcon Wing Door Technical Details

- Each door contains six inductive sensors, one capacitive sensor, and one sonar sensor
- Actuators: Electric motors with position feedback
- Opening time: Approximately 6–8 seconds fully deployed
- Integrated LED lighting in door panels

- Obstacle detection: Can detect ceiling height, nearby vehicles, and pedestrians; will adjust opening arc dynamically
- Winter operation: De-icing logic detects cold weather and applies increased force to break through ice seals (with warnings to user)

1.4 Model Y — Best-Selling Crossover SUV

Aspect	Details
Production Years	2020–present
Generations	Pre-Refresh (2020–2025), "Juniper" Refresh (2025–present)
Body Style	5-door compact crossover SUV, drag coefficient: 0.23 Cd
Dimensions	Length: 187.0 in, Width: 75.6 in (with mirrors folded), Height: 64.0 in, Wheelbase: 1 in
Curb Weight	RWD: 4,200 lbs (est.), Long Range AWD: 4,400 lbs, Performance: 4,500 lbs
Parts Commonality with Model 3	Approximately 75% shared components (reduced to ~65% with 4680 structural pac variants)

Variant Deep Dive

Model Y Rear-Wheel Drive

- **Battery:** LFP chemistry (most markets), 60–62 kWh, supplied by CATL or BYD (Blade Battery in some markets)
- **Range:** 260–320 miles (varies significantly by market and regulatory testing cycle)
- **0–60 mph:** 6.5–6.9 seconds (varies by battery source)
- **Availability:** Not offered in all markets; primary market is China, Europe, and select North American inventory

Model Y Long Range AWD

- **Battery Options:**
    - **2170 NCM Pack:** Used in Giga Berlin, Giga Shanghai, and Fremont-built vehicles; 75–82 kWh nominal

    - **4680 Structural Pack:** Used in Giga Texas vehicles; approximately 68–70 kWh nominal (lower capacity but reduced weight)

- **Range:** 310–330 miles (EPA estimates vary by battery source and wheel size)
- **0–60 mph:** 4.8 seconds (with 1-foot rollout)
- **Charging:** 250 kW peak (both battery types), 10–80% in 20–25 minutes

Model Y Performance AWD

- **Distinguishing Features:** 21" Überturbine wheels, lowered suspension (0.6" lower than Long Range), performance brake calipers (red), carbon fiber spoiler, aluminum alloy pedals
- **0–60 mph:** 3.5 seconds (with rollout)
- **Range:** 285–303 miles (EPA)
- **Top Speed:** 155 mph (software limited)
- **Track Mode:** Available on Performance models (added via OTA in 2023)

Structural Battery Pack (4680) Technical Details

- Introduced at Giga Texas (2022), later at Giga Berlin (limited)
- Battery cells are structural: Pack becomes part of vehicle frame, replacing the central floor section
- Cells are bonded into pack using polyurethane adhesive, no module structure (cell-to-pack design)
- Front and rear megacastings attach directly to structural battery pack
- Advantages: 10% reduction in mass, 30% reduction in parts count, improved torsional rigidity
- Disadvantages: Repairability complexity — battery pack removal requires specialized equipment, minor damage may require full pack replacement

Juniper Refresh (2025+) — Anticipated Changes

- Updated front and rear fascias (similar to Highland design language)
- Revised interior with ambient lighting, ventilated front seats
- Improved acoustic glass and noise insulation
- Potential range improvements via aerodynamics and thermal system updates
- HW4 (AI4) as standard across all variants

1.5 Cybertruck — Stainless Steel Pickup

Aspect	Details
Production Start	November 2023 (initial deliveries)
Production Ramp	2024–2025 (scaling to 250,000+ units annually)
Body Material	Ultra-hard 30X cold-rolled stainless steel (same alloy used in SpaceX Starship)
Exoskeleton Design	Body panels are structural; no separate frame rails under passenger compartment
Dimensions	Length: 223.7 in, Width: 86.6 in (with mirrors folded), Height: 70.5 in (adjustable air suspension), Wheelbase: 143.0 in
Curb Weight	AWD: 6,670 lbs, Cyberbeast: 6,890 lbs
Drag Coefficient	0.34 Cd (exceptional for a pickup truck)

Variant Deep Dive

Cybertruck All-Wheel Drive (AWD)

- **Powertrain:** Dual motor AWD (front and rear permanent magnet motors)
- **Horsepower:** 600 hp (estimated)
- **0–60 mph:** 4.1 seconds (with rollout)
- **Top Speed:** 112 mph (electronically limited)
- **Range:** 340 miles (EPA est. with all-terrain tires)
- **Towing Capacity:** 11,000 lbs
- **Payload Capacity:** 2,500 lbs
- **Battery:** 4680 cells, estimated 123 kWh usable capacity (unofficial)
- **Charging:** 800V architecture, V4 Supercharger optimized, peak rate 350 kW+ (350 kW–500 kW pending software and infrastructure rollout)
- **10–80% Charge Time:** Approximately 20–25 minutes on V4 Supercharger

Cybertruck Cyberbeast (Tri-Motor)

- **Powertrain:** Tri-motor AWD — one front permanent magnet, two rear permanent magnet motors with torque vectoring
- **Horsepower:** 845 hp (combined)
- **0–60 mph:** 2.6 seconds (with rollout) — "Beast Mode" launch control
- **Top Speed:** 130 mph
- **Range:** 320 miles (EPA est. with all-terrain tires)
- **Quarter Mile:** 11.0 seconds @ 119 mph (estimated)
- **Towing:** 11,000 lbs
- **Payload:** 2,500 lbs

Cybertruck Rear-Wheel Drive (Future)

- Announced but not yet in production (expected late 2025 or 2026)
- Estimated range: 250 miles
- Estimated price: $60,990 (pre-order pricing, subject to change)
- Battery: LFP chemistry likely

Key Technological Innovations

48-Volt Low-Voltage Architecture

- Industry standard is 12V for lights, windows, computers, and actuators
- 48V allows smaller gauge wiring (reduces copper usage by ~70%)
- Requires all components to be custom-designed (no off-the-shelf 12V parts)
- Enables higher-power features without current draw issues

Steer-by-Wire System

- No physical steering column; fully electronic connection between steering wheel and front/rear wheels
- Variable steering ratio: up to 10 degrees of rear wheel steering in opposite direction at low speeds (reduces turning circle to 43.5 ft — smaller than Model 3)
- At high speeds, rear wheels steer in same direction for stability
- Feedback is simulated; user can adjust steering feel weight

800V Charging Architecture

- Battery operates at nominal 800V (versus 400V in other Tesla models)
- Enables higher charging power without increasing current ($P = V \times I$)

- V4 Superchargers can deliver up to 1,000V and 615A (615 kW theoretical max, currently software limited)
- Allows Cybertruck to add up to 136 miles of range in 15 minutes under optimal conditions

Armor Glass

- Thickened laminated glass developed in-house
- Withstands hail impacts and attempted break-ins
- Note: The 2019 demonstration breakage was due to pre-existing stress fractures from prior sledgehammer tests; production glass meets specifications

Vault (Truck Bed)

- Length: 6 ft (72 in), Width: 51 in (at floor), Depth: 21 in
- Powered roll-top tonneau cover (integrated into sail pillars)
- 120V and 240V outlets (up to 9.6 kW power export capability)
- Total bed volume with tonneau closed: 67 cu ft (lockable and weather-sealed)

1.6 Tesla Semi — Class 8 Electric Truck

Aspect	Details
Production Start	2022 (limited), 2025 (volume ramp)
Configuration	Class 8 semi-truck, 500-mile and 300-mile range variants
Powertrain	Four independent motors (one per rear wheel), total power output estimated 1,000+ hp
0–60 mph (Loaded)	20 seconds with 82,000 lb gross vehicle weight (GVW)
5% Grade Speed	65 mph with full load (industry average is 25–35 mph)
Charging	Megacharger (1 MW+), 500 miles of range added in approximately 30 minutes
Battery	Estimated 900–1,000 kWh capacity, 4680 cells, structural pack

Aspect	Details
Efficiency	< 2 kWh per mile (versus 6–8 MPG diesel equivalent)
Production Site	Giga Nevada (expanded facility)

1.7 Tesla Roadster (Gen 2) — Future Production

Aspect	Details
Announced	2017, production delayed (current target: 2026–2027)
Configuration	4-seat coupe (2+2 seating), removable glass roof
Powertrain	Tri-motor AWD, "SpaceX Package" optional with cold air thrusters
0–60 mph	1.1 seconds (with SpaceX package, claimed)
Top Speed	250+ mph (claimed)
Range	620 miles (claimed)
Base Price	$200,000 (initial reservation), $250,000 (Founders Series)

SECTION 2: BATTERY TECHNOLOGY — COMPREHENSIVE TECHNICAL REFERENCE

2.1 Battery Cell Form Factors and Chemistry Comparison

Cell Type	Dimensions	Chemistry	Capacity	Applications	Key Characteristics
18650	18mm × 65mm	NCA (Nickel-	3.0–3.5 Ah	Model S, Model X (pre-2021)	Original Tesla cell, thousands per pack (7,104 in 100 kWh

Cell Type	Dimensions	Chemistry	Capacity	Applications	Key Characteristics
		Cobalt-Aluminum)			pack), proven longevity
2170	21mm × 70mm	NCM 811, NCA variants	4.5–5.0 Ah	Model 3, Model Y (most), Model S/X (2021+), Semi	Higher energy density than 18650, optimized cooling, manufactured at Giga Nevada
4680	46mm × 80mm	NCM 955 (high nickel), future variants	26–30 Ah	Cybertruck, Model Y (Giga Texas), Semi	Tabless electrode, 5x energy per cell, 6x power, enables structural packs
LFP (Prismatic)	Varies (blade or pouch)	Lithium Iron Phosphate	60–62 Ah (pack level)	Model 3 RWD, Model Y RWD (select markets)	No cobalt, lower cost, longer cycle life, lower energy density

2.2 Battery Chemistry Deep Dive

NCA (Nickel-Cobalt-Aluminum)

- Composition: Approximately 80% nickel, 15% cobalt, 5% aluminum
- Advantages: High energy density (250–260 Wh/kg), excellent power delivery
- Disadvantages: Cobalt supply chain concerns, thermal stability requires robust BMS
- Used in: Pre-2023 Model S/X, some 2170 cells

NCM 811 (Nickel-Cobalt-Manganese)

- Composition: 80% nickel, 10% cobalt, 10% manganese (811 designation)
- Advantages: Balance of energy density (260–270 Wh/kg), power, and cost
- Used in: Most 2170 cells in Model 3/Y (2021+), some 4680 variants

LFP (Lithium Iron Phosphate)

- Voltage: 3.2V nominal (versus 3.6–3.7V for NCA/NCM)
- Energy Density: 160–180 Wh/kg (pack level lower due to lower cell voltage)
- Cycle Life: 2,000–4,000 cycles to 80% capacity (versus 1,000–1,500 for NCA/NCM)
- Thermal Runaway Temperature: >270°C (versus ~210°C for NCA/NCM)
- Calendar Life: 15–20 years in moderate climates
- Charging: Can accept 1C+ charge rates, but peak voltage is lower, limiting peak power at Superchargers
- **Critical Usage Note:** LFP batteries require regular 100% charging (at least once per week) for BMS calibration due to flat voltage curve (voltage doesn't correlate well with state of charge)

4680 Chemistry (Current and Future)

- Current (Gen 1): High-nickel NCM 955 (90% nickel, 5% cobalt, 5% manganese) — "dry battery electrode" process not yet fully implemented
- Future (Gen 2–3): Dry battery electrode (DBE) manufacturing, reducing production cost by ~50% and energy consumption by ~70%
- Energy density target: 300 Wh/kg (cell level), 200+ Wh/kg (pack level)

2.3 Battery Pack Architecture Evolution

Generation	Vehicles	Configuration	Key Features
Gen 1 (18650)	Model S/X (2012–2020)	Modular, 16 modules, liquid cooling between modules	Serviceable modules, active thermal management, 400V architecture
Gen 2 (2170)	Model 3/Y (2017–2022, non-structural)	4 modules in skateboard configuration, integrated cooling channels	Reduced complexity, improved thermal uniformity
Gen 3 (Structural 2170)	Model Y (Giga Berlin, early Giga Texas)	Cell-to-pack without upper modules, cells bonded to upper chassis	First step toward structural integration, 10% weight reduction
Gen 4 (Structural 4680)	Cybertruck, Model Y (Giga Texas)	Full structural, cells bonded between castings, no separate pack tray	Maximum integration, 30% parts reduction, 800V architecture (Cybertruck)

2.4 Thermal Management System

Octovalve (Model 3/Y)

- Centralized thermal management manifold with 8 ports
- Manages heat flow between battery, drive units, cabin HVAC, and radiator
- Enables heat pump operation (introduced 2020 in Model Y, 2021 in Model 3)
- **Superbottle** (earlier Model 3): Integrated coolant pump, reservoir, and valve assembly

Heat Pump (All Vehicles 2021+)

- Coefficient of Performance (COP): 2–4 (versus 1 for resistive heating)
- Operates down to -30°C, with efficiency dropping as temperature drops
- Sources waste heat from battery and drive units to warm cabin
- In extreme cold, augments with resistive heating elements (PTC heaters)

Preconditioning Logic

- When navigating to Supercharger, vehicle calculates optimal battery temperature
- Target temperature: 40–50°C (104–122°F) for 250 kW charging
- Preconditioning begins 15–30 minutes before arrival depending on ambient temperature
- Energy cost: Typically 2–5% of battery capacity consumed for heating/cooling during preconditioning

2.5 Battery Degradation and Longevity

Fleet Data (Based on 10,000+ vehicles tracked by third-party services)

- Average degradation after 50,000 miles: 5–6%
- Average degradation after 100,000 miles: 8–10%
- Average degradation after 200,000 miles: 10–12%
- Model 3/Y (2170) shows slightly lower degradation than Model S/X (18650) due to improved thermal management

Factors Affecting Degradation

1. **Charging Habits:** Frequent Supercharging (100% DC fast charging) increases degradation by ~1–2% over 50,000 miles compared to primarily AC home charging
2. **State of Charge (SoC):** Time spent at >90% SoC or <10% SoC accelerates calendar aging
3. **Temperature:** High ambient temperatures (>40°C/104°F) increase degradation rates
4. **Depth of Discharge:** Shallow cycles (e.g., 50–70% SoC) are gentler than deep cycles (10–90% SoC)

LFP Battery Specifics

- Calendar aging: Lower than NCA/NCM; estimated 70–80% capacity after 20 years
- Cycle aging: 2,000–4,000 cycles to 80% capacity
- Charging: Regular 100% charging does NOT accelerate degradation (unlike NCA/NCM)
- BMS calibration: Requires periodic full charge to 100% and discharge to <10% for accurate range estimation

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SECTION 3: CHARGING INFRASTRUCTURE — COMPLETE REFERENCE

3.1 North American Charging Standard (NACS)

Specification	Details
Physical Connector	Single unified connector for AC and DC, 2.5 inches in diameter, 5 pins (2 power, 2 signal, 1 ground)
AC Charging	Up to 19.2 kW (80 amps at 240V) — requires Gen 3 Wall Connector and vehicle compatibilit
DC Charging (V3)	Up to 250 kW (500A at 500V)
DC Charging (V4)	Up to 350 kW (615A at up to 1000V)
Communication Protocol	CCS (Combined Charging System) protocol over PLC (Power Line Communication)
Adoption	Ford, GM, Rivian, Volvo, Polaris, Mercedes, Nissan, and others announced NACS adoption (2024–2026 transition)

3.2 Supercharger Network Specifications

Generation	V2	V3	V4
Max Power	150 kW (shared between two stalls)	250 kW (dedicated per stall)	350 kW (software limited, capable of 500–600 kW)

Generation	V2	V3	V4
Voltage Range	Up to 500V	Up to 500V	Up to 1000V
Max Current	350A (shared)	600A	615A
Cable Cooling	No	Liquid-cooled	Liquid-cooled, longer cables
Deployment Year	2014–2019	2019–present	2023–present
Non-Tesla Support	No (Tesla-only connector, older protocol)	Yes (with adapter or NACS native)	Yes (native CCS support, longer cables)
Stalls per Site	8–12 typical	8–40 typical	8–96 typical (megasites)

Charge Curve Example (Model Y Long Range, 250 kW V3)

- 5% SoC: 250 kW peak
- 10–30% SoC: 230–250 kW sustained
- 40% SoC: 180 kW
- 50% SoC: 150 kW
- 60% SoC: 120 kW
- 70% SoC: 90 kW
- 80% SoC: 60 kW
- 80–100% SoC: 15–60 kW (tapering continuously)

Time to Charge (250 kW vehicle)

- 10–50%: 10–12 minutes
- 10–80%: 22–25 minutes
- 10–100%: 45–55 minutes

3.3 Home Charging Options

Charger	Power	Installation Requirements	Cost (Equipment)	Charging Speed (miles/hour)
Mobile Connector (120V)	1.3 kW (12A)	Standard household outlet	$250	3–4
Mobile Connector (240V NEMA 14-50)	7.7 kW (32A)	50A breaker, 14-50 outlet	$250 + outlet install	20–30
Wall Connector (Gen 3)	11.5 kW (48A)	60A breaker, hardwired	$475	30–44
Wall Connector (Universal)	11.5 kW (48A)	60A breaker, hardwired	$550	30–44 (includes built-in J1772 adapter)
Third-Party (ChargePoint, etc.)	9.6–11.5 kW (40–48A)	50–60A breaker	$500–$800	25–40

3.4 Destination Charging and Third-Party Networks

Tesla Destination Charging

- Network of Level 2 chargers at hotels, restaurants, and commercial locations
- Typically 8–16 kW (208–240V, 32–80A)
- Free to use for Tesla owners (host sets terms)
- Over 35,000 connectors globally

Third-Party Networks (with NACS Adapters)

- **CCS Adapter (Tesla OEM):** $250–$300, enables up to 250 kW at CCS stations (Electrify America, EVgo, etc.)
- **CHAdeMO Adapter:** Discontinued, limited to 50 kW
- **Non-Tesla Supercharging:** Select V3/V4 sites open to non-Tesla EVs with adapter or native NACS

SECTION 4: AUTOPILOT AND FULL SELF-DRIVING — TECHNICAL ARCHITECTURE

4.1 Hardware Generations

Component	HW2.5 (2017–2019)	HW3 (2019–2024)	HW4 / AI4 (2024–present)
Autopilot Computer	NVIDIA Tegra + Infineon (2×)	Tesla FSD Computer (14 nm Samsung)	Tesla AI4 Computer (7 nm)
Neural Network Performance	110 TOPS (trillion ops/sec)	2,300 TOPS (144 TOPS per chip, 2 chips)	3,000–5,000 TOPS (estimated)
Cameras	8× 1.2 MP (HDR)	8× 1.2 MP (HDR)	12× 5 MP (increased resolution, dynamic range)
Radar	Front-facing radar (Bosch)	Front-facing radar (HW2.5), removed in Tesla Vision (2021+)	Optional HD Radar (Phoenix) in some markets, not used for autonomy
Ultrasonic Sensors	12× (front/rear/sides)	12× (removed 2022–2023)	None (Tesla Vision replaces)
Steering Wheel Torque Sensor	Yes	Yes	Capacitive touch (hands-on detection)

4.2 Camera Configuration (HW4)

Position	Resolution	Field of View	Function
Main Front (Narrow)	5 MP	35°	Long-range object detection, traffic signals
Main Front (Wide)	5 MP	120°	Intersection visibility, cross-traffic
Main Front (Fisheye)	5 MP	180°	Close-range, traffic lights, overhead signage
B-Pillar Left/Right	5 MP	80°	Adjacent lane traffic, merges

Position	Resolution	Field of View	Function
Fender Left/Right (Repeaters)	5 MP	90°	Rearward lane changes, blind spots
Rear (Trunk)	5 MP	110°	Rearward driving, backup
Front Bumper (Cybertruck only)	5 MP	45°	Off-road visibility, front blind spot

4.3 FSD Software Stack (Version History)

Version	Architecture	Key Features	Release
FSD v10	Legacy C++ codebase, rule-based planning	Highway Autopilot, Navigate on Autopilot	2021
FSD v11	Unified stack (city + highway)	Single stack for all driving, improved city streets	2022–2023
FSD v12	End-to-end neural networks (AI-driven)	Removed 300,000+ lines of C++ code, "photon to control"	2024–present
FSD v12.5+	Highway optimization, improved AI inference	Reduced interventions, more natural driving	2025+

FSD (Supervised) Capabilities (Current)

- City street navigation (intersections, traffic lights, stop signs, roundabouts)
- Unprotected left and right turns
- Lane selection and navigation based on route
- Speed adjustments based on road conditions and speed limits
- Obstacle avoidance (construction cones, debris, parked vehicles)
- Autopark (parallel, perpendicular, pull-in)
- Summon and Smart Summon (parking lot navigation to driver)

Limitations (Current)

- Requires active driver supervision (hands on steering wheel, eyes on road)
- Performance degraded in heavy rain, snow, fog
- Unprotected left turns remain challenging
- Construction zones require driver takeover
- Not certified for unsupervised autonomous driving in any jurisdiction

4.4 Safety Systems

Active Safety Features (Standard on All Vehicles)

- Automatic Emergency Braking (AEB): Operates up to 90 mph, pedestrian and cyclist detection
- Lane Departure Avoidance: Steering intervention if crossing lane lines without turn signal
- Obstacle-Aware Acceleration: Limits acceleration if obstacle detected ahead
- Side Collision Avoidance: Steering correction if vehicle drifts into adjacent lane with oncoming traffic
- Emergency Lane Departure Avoidance: Combines lane departure and side collision avoidance

Sentry Mode

- Uses exterior cameras to record when vehicle detects threat
- Records all four sides simultaneously
- Records to USB drive (requires formatted drive in glovebox)
- Mobile app notifications for significant events
- Power consumption: Approximately 200–250 watts (2–3% battery loss per 10 hours)

Dashcam

- Continuous recording from front, side, and rear cameras
- Saves last 10 minutes on demand or automatically on collision detection
- Requires USB drive (Tesla recommends high-endurance drives)

SECTION 5: MAINTENANCE — COMPREHENSIVE GUIDE

5.1 Recommended Service Intervals

Service Item	Interval	Notes
Tire Rotation	6,250 miles (10,000 km)	Required for even wear; more frequent for Performance mode
Cabin Air Filter	2 years (Model 3/Y), 3 years (S/X)	HEPA filter (Biohazard Defense) requires more frequent replacement in poor air quality regions
Brake Fluid Test	2 years	Replace if moisture content exceeds 3%
A/C Desiccant Bag	4 years (Model 3), 6 years (S/X/Y)	Maintains A/C efficiency; replacement recommended
Battery Coolant	4 years (Model 3/Y), 8 years (S/X)	Performed by Tesla service only
Brake Caliper Lubrication	1 year (cold climates)	Required if roads salted in winter
Wiper Blades	6–12 months	Replace when streaking or chattering
High Voltage Battery	No scheduled maintenance	Sealed system, no user-serviceable components

5.2 Common Issues by Model

Model S/X (Pre-2021)

- **Door Handle Failure:** Retractable handle microswitches fail ($200–400 repair per door)
- **MCU (Media Control Unit) eMMC Failure:** 8GB eMMC memory wears out, causing touchscreen failure (covered by recall/warranty for certain VINs)
- **Air Suspension Compressor:** Failure due to moisture ingress ($1,200–1,500 repair)
- **Rear Drive Unit (Large Drive Unit):** Coolant seal failure leading to electrical failure (early models, ~3% failure rate by 100,000 miles)

Model 3/Y (Common)

- **Front Upper Control Arm:** Ball joint wear causing squeaking noise (typically 30,000–50,000 miles). Repair cost: $200–400 per side if out of warranty.

- **Rear Glass (Model Y):** Stress cracks near trunk hinge (covered under warranty if no impact evidence)
- **Paint Quality:** Early models had thin paint; improved with Giga Shanghai and Giga Berlin production
- **Heat Pump Noise:** Operational noise normal; loud rattling may indicate foreign object or mounting issue

Cybertruck (Emerging)

- Panel gaps and alignment (early production, improving with ramp)
- Tonneau cover operation (initial issues with debris ingress)
- Touchscreen responsiveness (software updates addressing)

5.3 Warranty Coverage

Component	Warranty Period	Coverage
New Vehicle Limited Warranty	4 years or 50,000 miles	Bumper-to-bumper (exclu wear items)
Battery and Drive Unit (Model 3/Y RWD)	8 years or 100,000 miles	Minimum 70% capacity retention
Battery and Drive Unit (Model 3/Y Long Range/Performance)	8 years or 120,000 miles	Minimum 70% capacity retention
Battery and Drive Unit (Model S/X)	8 years or 150,000 miles	Minimum 70% capacity retention
Battery and Drive Unit (Cybertruck)	8 years or 150,000 miles	Minimum 70% capacity retention
Corrosion Perforation	12 years, unlimited mileage	Rust-through of body pan
Seat Belts and Airbags	5 years or 60,000 miles (seat belts), 10 years (airbags)	Safety restraint systems

5.4 Cost of Ownership Comparison (5 Years / 75,000 Miles)

Expense Category	Tesla Model 3 (EV)	BMW 3 Series (ICE)
Fuel/Energy	$4,500 (home charging at $0.15/kWh, 3.5 mi/kWh)	$10,500 (gas at $3.50/gal, 25 mp
Maintenance	$1,500 (tires, filters, brake fluid)	$5,000 (oil changes, brakes, belts fluids)
Repairs (Out of Warranty)	$500 (average minor repairs)	$1,500 (average minor repairs)
Total (excluding depreciation)	$6,500	$17,000

SECTION 6: SOFTWARE AND CONNECTIVITY

6.1 Connectivity Tiers

Feature	Standard Connectivity (Free)	Premium Connectivity ($9.99/mo or $99/yr)
Navigation	Yes (with traffic data)	Yes (with live traffic visualization)
Satellite Maps	No	Yes
Music Streaming (Slacker/Spotify)	No (requires Bluetooth or USB)	Yes (over cellular)
Video Streaming (Netflix, YouTube, etc.)	Wi-Fi only	Yes (over cellular)
Live Sentry Mode Camera View	No	Yes (via mobile app)
Internet Browser	Wi-Fi only	Yes (over cellular)
Caraoke	Wi-Fi only	Yes

Feature	Standard Connectivity (Free)	Premium Connectivity ($9.99/mo or $99/yr)
Traffic Light/Stop Sign Visualization	No (requires Premium)	Yes

6.2 OTA Update History (Major Features Added)

Year	Feature	Impact
2012	Initial Model S software	First connected car interface
2014	Autopilot (HW1)	First production autonomy features
2015	Ludicrous Mode (Model S)	Over-the-air performance upgrade
2017	Enhanced Autopilot (HW2)	Navigate on Autopilot introduced
2018	Dog Mode, Sentry Mode	Cabin temperature maintenance, security recording
2019	Camp Mode, Smart Summon	Climate control while parked, parking lot navigation
2020	Heat Pump optimization (Model Y)	Improved cold weather efficiency
2021	Tesla Vision (Vision-only)	Removed radar dependency
2022	Steam Gaming (Model S/X)	AAA gaming integration
2023	Apple Music, Zoom, Light Show	Expanded entertainment
2024	FSD v12 (End-to-End AI)	Removed 300,000 lines of C++ code
2025	Adaptive Suspension improvements	Real-time damping adjustments based on road scanning

6.3 Mobile App Features

Core Functions

- Vehicle location and status
- Climate control (preconditioning, Dog Mode, Camp Mode)
- Charge management (start/stop, set limits, schedule)
- Keyless driving (Phone Key)
- Remote window and trunk control
- Service scheduling and roadside assistance

Premium Features

- Live camera feed (Sentry Mode)
- Tire pressure monitoring
- Charge history and statistics
- Solar integration (Powerwall, Solar Roof)

SECTION 7: MANUFACTURING AND GIGAFACTORIES

7.1 Factory Overview

Factory	Location	Production Start	Models	Annual Capacity (2024 est.)	Key Technology
Giga Shanghai	Shanghai, China	2019	Model 3, Model Y	950,000+	Highest output, lowest cost
Giga Berlin	Grünheide, Germany	2022	Model Y	250,000 (scaling to 500,000)	Advanced paint shop, structural packs
Giga Texas	Austin, Texas, USA	2022	Model Y, Cybertruck	375,000+ (scaling to 500,000+)	4680 cells, Cybertruck production
Giga Nevada	Sparks, Nevada, USA	2014	2170 cells, Semi, powertrain	30+ GWh battery capacity	Battery cell manufacturing

Factory	Location	Production Start	Models	Annual Capacity (2024 est.)	Key Technology
Fremont Factory	Fremont, California, USA	2010	Model S, X, 3, Y	600,000+	Legacy assembly, vehicle finalization

7.2 Megacasting Technology

Giga Press Specifications

- **Supplier:** Idra Group (Italy)
- **Clamping Force:** 6,000–9,000 tons
- **Shot Weight:** 80–100 kg (176–220 lbs) of molten aluminum per cycle
- **Cycle Time:** 90–120 seconds per casting

Part Consolidation

Component	Traditional (Stamped)	Megacast (Integrated)
Rear Underbody	70–80 stamped parts, 300–400 welds	1–2 cast parts, no welding
Front Underbody	50–60 parts	1–2 cast parts
Weight Reduction	Baseline	10–15% lighter
Cost Reduction	Baseline	20–30% reduction

7.3 Supply Chain and Vertical Integration

Tesla Manufactured Components

- Battery cells (2170 at Giga Nevada, 4680 at Giga Texas)
- Drive units (motors and inverters) — all factories
- Seats (Giga Shanghai, Giga Berlin)
- Castings (in-house Giga Presses)
- Autopilot/FSD computers (in-house design, outsourced fabrication)

- Power electronics (silicon carbide inverters)

Key Suppliers

- **Battery Cells:** Panasonic (2170 at Giga Nevada), LG Energy Solutions (2170 for China), CATL (LFP), BYD (Blade LFP)
- **Semiconductors:** Infineon (IGBTs, SiC), STMicroelectronics (SiC)
- **Glass:** AGC, Saint-Gobain
- **Tires:** Michelin, Continental, Pirelli, Hankook

SECTION 8: FUTURE ROADMAP AND UPCOMING TECHNOLOGIES

8.1 Confirmed Upcoming Models

Model	Target Launch	Platform	Notes
"Model 2" / Compact Vehicle	2025–2026	Next-gen platform	$25,000–$30,000 target price, 300+ mile range, manufactured at Giga Mexico and Giga Shanghai
Tesla Roadster (Gen 2)	2026–2027	Modified Model S platform	SpaceX package optional, 0–60 in 1.1 sec
Cybertruck RWD	2025–2026	Cybertruck platform	Lower-cost variant, LFP battery
Model Y Juniper	2025	Model Y platform	Refresh with Highland-style updates
Tesla Semi (Volume)	2025–2026	Semi platform	500-mile range variant scaling

8.2 Technology Roadmap

Battery Technology

- **Dry Battery Electrode (DBE):** Reduce manufacturing cost by 50%, energy consumption by 70%, enable 300+ Wh/kg cells

- **Sodium-Ion Batteries:** Potential for stationary storage and low-cost entry vehicles (no cobalt, no lithium)
- **1,000+ Wh/kg Cells (Long-term):** Lithium-metal and solid-state research

Autonomy

- **Unsupervised FSD:** Regulatory approvals targeted for select markets (initially California, Texas) in 2026–2027
- **Cybercab (Robotaxi):** Dedicated autonomous vehicle with no steering wheel or pedals, target production 2026–2027
- **Robovan:** Autonomous people/ goods mover

Charging Infrastructure

- **V4 Supercharger Network:** 350–500 kW, 1,000V architecture, 60,000+ stalls globally by 2026
- **Wireless Charging:** Inductive charging pads for home and commercial use (prototype stage)
- **Megacharger Network:** 1 MW+ for Semi, 1,800+ stalls by 2026

Manufacturing

- **Unboxed Process (Next-Gen):** Modular assembly process reduces factory footprint by 40%, cost by 50%
- **Giga Mexico:** First factory designed for Unboxed process, production start 2026

SECTION 9: REGULATORY AND COMPLIANCE INFORMATION

9.1 Tax Credits and Incentives (US)

Federal Tax Credit (IRA, 2024–2032)

- Up to $7,500 for qualifying vehicles
- **Criteria:**
    - Vehicle MSRP < $80,000 (SUV/truck) or < $55,000 (sedan)
    - Buyer income limits: $300,000 joint, $225,000 head of household, $150,000 single
    - Battery component sourcing and critical mineral requirements
- **Tesla Models Currently Eligible:** Model 3 RWD, Model 3 Long Range, Model Y RWD, Model Y Long Range, Model Y Performance (varies by configuration and manufacturing location)

State-Level Incentives

- California: CVRP up to $2,000 (income-based)

- Colorado: Up to $5,000
- Massachusetts: Up to $3,500
- New York: Up to $2,000
- Various utility incentives for home charger installation

9.2 Safety Ratings

Model	NHTSA Overall Rating	Euro NCAP	IIHS
Model 3	5-star (all categories)	5-star	Top Safety Pick+
Model Y	5-star (all categories)	5-star	Top Safety Pick+
Model S	5-star (all categories)	5-star (2014 rating)	Not rated (older model)
Model X	5-star (all categories)	5-star	Top Safety Pick+ (2022)
Cybertruck	Pending	N/A	N/A

9.3 Recalls and Campaigns (Notable)

Issue	Models Affected	Resolution
MCU eMMC Failure (Recall)	Model S/X (2012–2018)	Replacement 64GB eMMC or upgraded MCU2 (paid upgr option)
Rearview Camera Cable	Model 3 (2017–2020)	Cable replacement
Bolt Torque (Steering Rack)	Model Y (2020–2021)	Bolt inspection and torque
FSD Beta Recall (NHTSA)	FSD-equipped vehicles (2023)	OTA update to improve rolling stop detection

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SECTION 10: ENVIRONMENTAL IMPACT AND SUSTAINABILITY

10.1 Lifecycle Emissions

Vehicle	Manufacturing Emissions	Use-Phase Emissions (150,000 miles)	Total Lifecycle
Model 3 (US grid mix)	12–15 tons CO2e	20–25 tons CO2e	32–40 tons CO2e
Model 3 (renewable charging)	12–15 tons CO2e	5–8 tons CO2e	17–23 tons CO2e
ICE Sedan (30 mpg)	8–10 tons CO2e	60–70 tons CO2e	68–80 tons CO2e

CO2e = Carbon dioxide equivalent

10.2 Recycling and Circular Economy

Battery Recycling (Giga Nevada)

- Tesla operates in-house battery recycling facility
- Achieves 92% material recovery rate (lithium, cobalt, nickel, copper, aluminum)
- Closed-loop system: Recycled materials return to cell manufacturing

Manufacturing Sustainability

- Giga Berlin: Water consumption 1.8 L per vehicle (industry average: 4–6 L)
- Giga Shanghai: Zero wastewater discharge
- Solar installations on factory roofs (Giga Nevada, Giga Shanghai, Giga Texas)

USER QUESTIONS FOR RAG TESTING (EXPANDED SET)

Simple Retrieval (Factoid)

1. What is the difference in charging chemistry, peak Supercharger rate, and recommended daily charging limit between a Model 3 Long Range (NCM) and a Model 3 Rear-Wheel Drive (LFP)?
2. Which Tesla models currently utilize the 4680 battery cell form factor, and what is the manufacturing advantage of the tabless electrode design over traditional 2170 cells?
3. What are the towing capacity, payload capacity, 0–60 mph time, and peak charging power specifications for the Cybertruck Cyberbeast?

4. What is the difference between Hardware 3 (HW3) and Hardware 4 (HW4) in terms of camera resolution, processing power (TOPS), and sensor suite?
5. What is the Octovalve, and how does it function within the Model Y thermal management system to improve efficiency in cold weather?

Multi-Hop Reasoning (Requires Combining Sections)

6. I drive a 2022 Model Y Long Range with a 2170 NCM battery. I am planning a road trip from Texas to Colorado in winter. Based on the battery chemistry, thermal management system, and BMS functionality, what steps should the vehicle take automatically to ensure optimal charging speeds at V3 Superchargers? How would this differ if I had a Model Y RWD with LFP battery instead?
7. A user has a 2024 Model S Plaid with HW4 and wants to use FSD (Supervised) in a city with heavy snow. Considering the vision-only architecture and the removal of ultrasonic sensors, what potential limitations might the system face that a traditional radar-based or sensor-fusion system would handle differently? Reference specific camera limitations.
8. Compare the structural battery pack used in the Model Y produced at Giga Texas (4680) to the modular 2170 pack used in Model Y produced at Giga Berlin. Analyze the differences in repairability, mass, structural rigidity, and charging performance.
9. A user owns a 2015 Model S 85D and is considering upgrading to a Cybertruck. What are the key changes they will experience regarding low-voltage electrical architecture (12V vs. 48V), steering mechanism (rack-and-pinion vs. steer-by-wire), charging connector (proprietary Tesla vs. NACS), and battery chemistry? Will their existing Gen 2 Wall Connector work with the new vehicle without modification?
10. Based on fleet degradation data, what is the expected battery degradation for a Model 3 Long Range after 100,000 miles if the owner primarily uses Supercharging (80% DC fast charging) versus a user who primarily charges at home (95% AC Level 2)? Include the impact of charging habits on NCM vs. LFP batteries.

Comparative Analysis (Requires Synthesis)

11. Compare the maintenance schedule and expected 5-year cost of ownership of a 2024 Model 3 Performance to a 2024 BMW M340i (ICE). Focus specifically on brake wear, fluid replacement cycles, tire rotation intervals, and fuel/energy costs using US national averages.
12. Analyze the differences between the V3 and V4 Supercharger networks. Include cable cooling, voltage architecture, peak power, compatibility with 800V vehicles (Cybertruck), and deployment timeline. What infrastructure changes are required for a V4 site versus a V3 site?
13. Compare the Autopilot hardware evolution from HW2.5 to HW3 to HW4. Include TOPS (processing power), camera resolution, sensor redundancy, and the transition to Tesla Vision. How has the removal of radar and USS impacted functionality in low-visibility conditions?

Hypothetical / Scenario-Based

14. I have a Model X Plaid with Falcon Wing doors and live in a dusty, high-salt winter climate. I am concerned about the long-term reliability of the door mechanisms. What are the known common issues with the Falcon Wing door system (sensors, actuators, seals), and what warranty protections typically cover these mechanisms? What preventive maintenance is recommended?

15. A user experiences a front suspension squeaking noise on their 2020 Model 3 at 35,000 miles. Based on common issues documented in the maintenance corpus, what is the most likely failed component? What is the typical repair cost out of warranty, and what is the safety risk if left unaddressed?

16. A user with a new Model Y RWD (LFP) is attempting to charge to 100% at a V3 Supercharger for a road trip. They notice that charging speed drops significantly after 80%, adding 30 minutes to reach 100%. Is this a hardware defect, or does the BMS logic treat LFP and NCA packs differently regarding charge curves? Explain the technical reason for the taper difference.

17. A Cybertruck owner wants to use the vehicle to power their home during a power outage. What is the maximum power export capability via the Vault 240V outlet, and what equipment would be required to connect this to a residential electrical panel? Compare this to the Powerwall or Ford F-150 Lightning's Home Integration System.

Edge Cases / Troubleshooting

18. My 2018 Model S touchscreen has become slow, unresponsive, and the backup camera sometimes doesn't display. Based on known issues, what is the most likely root cause? What recall or warranty coverage exists for this issue, and what are my repair options (eMMC replacement vs. MCU2 upgrade)?

19. I live in an apartment and cannot install a home charger. I rely on a V3 Supercharger 5 miles away. I have a Model 3 Long Range (NCM). What is the long-term impact on battery health if I exclusively use DC fast charging (Supercharging) for 50,000 miles? Provide quantitative degradation estimates based on fleet data.

20. During a software update, my Model Y displayed an error message and required service center intervention. What are the common causes of failed OTA updates, and what redundancy mechanisms exist in the vehicle architecture to prevent bricking? Can the vehicle recover without a service visit?

Manufacturing & Logistics

21. I live in Europe and want to purchase a Model Y with the 4680 structural battery pack. Which Gigafactory supplies vehicles to my market, and is the 4680 variant available in Europe, or is it specific to the North American market? If not, what battery technology will my European Model Y contain?

22. Explain the megacasting process used at Giga Texas. What are the part consolidation numbers (stamped parts replaced, weld reduction) and the claimed cost and weight savings for the rear underbody? How does this impact repairability after a collision?

23. A user is deciding between a Model Y from Giga Shanghai vs. Giga Berlin. What are the differences in paint quality, battery chemistry (LFP vs. NCM), and observed build quality consistency between these factories based on owner-reported data?

Future & Technology Roadmap

24. What is the "Unboxed Process" announced for next-generation Tesla vehicles, and how does it differ from traditional automotive assembly lines? What cost and factory footprint reductions are claimed, and at which factory will it first be deployed?
25. Based on the technology roadmap, what battery chemistries and form factors are expected to enter production by 2027? What are the target energy densities (Wh/kg) and cost reductions ($/kWh) for the Dry Battery Electrode process and potential sodium-ion cells?
26. When is unsupervised FSD (Robotaxi capability) currently targeted for regulatory approval, and in which markets? What hardware is required for full autonomy (HW3, HW4, or future HW5), and what is the estimated timeline for deployment?

End of Expanded Information Corpus

This corpus now contains approximately **8–10 dense pages** of structured, factual information across 10 sections with 40+ subsections, followed by **26 testing questions** ranging from simple retrieval to multi-hop reasoning, comparative analysis, and future roadmap inquiries. The content is designed to serve as a high-quality knowledge base for a RAG system with rich metadata potential.